

Study Guide: Chapter 16 – Capacitors

Concept Mastery & Problem Solving

1 The Hydraulic Analogy

To visualize how a capacitor works, compare it to a water tank:

Hydraulics vs. Electricity

- **Water Pressure (h)** $\leftrightarrow$ **Voltage (U)**
- **Quantity of Water** $\leftrightarrow$ **Electric Charge (Q)**
- **Tank Surface Area** $\leftrightarrow$ **Capacitance (C)**

Insight: A wider tank (larger C) stores more water (charge Q) for the same water level (voltage U).

2 Physical Principles: The Parallel Plate

The capacitance C is determined solely by the **geometry** and the **dielectric**:

- **Area (S)**: Doubling the plate area **doubles** the capacitance.
- **Separation (d)**: Doubling the distance **halves** the capacitance.
- **Permittivity (ϵ)**: Inserting a dielectric like Mica ($\epsilon_r = 5.4$) multiplies the capacitance by 5.4 compared to air ($C = \epsilon_r C_0$).

3 Conceptual Check: RC Dynamics

1. **At the instant charging begins ($t = 0$)**: Why is the current at its maximum? (Answer: Because the capacitor is empty, $u_C = 0$, so the full voltage E is across the resistor R).
2. **At steady state ($t = 5\tau$)**: Why does the current stop? (Answer: Because $u_C = E$, opposing the source perfectly. The capacitor acts as an **open circuit**).
3. **The Time Constant (τ)**: How can you make a capacitor charge faster? (Answer: Decrease R or decrease C).

4 Critical Comparison: Series vs. Parallel

Property	Parallel Grouping	Series Grouping
Total Capacity	Increases ($C_{eq} > \text{any } C_i$)	Decreases ($C_{eq} < \text{any } C_i$)
Voltage (U)	Same across all	Divided among them
Charge (Q)	Divided among them	Same on all plates
Analogy	Joining tanks side-by-side	Increasing plate separation

5 Safety and Energy

- **Stored Energy:** $W = \frac{1}{2}CU^2$. Notice that energy is proportional to the *square* of the voltage.
- **Practical Warning:** High-voltage capacitors can hold a lethal charge long after power is disconnected. Always discharge them through a resistor before handling.

Use this guide alongside the Chapter Summary to prepare for problem-solving apps.